IF69D-S11 e OPEET014-OPEL

Processamento Digital de Imagens

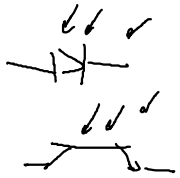
Aula 01(b)

Universidade Tecnológica Federal do Paraná - UTFPR - Curitiba
Departamento Acadêmico de Eletrônica - DAELN
Curso de Engenharia Eletrônica
Prof. Humberto R. Gamba e Prof. Gustavo B. Borba

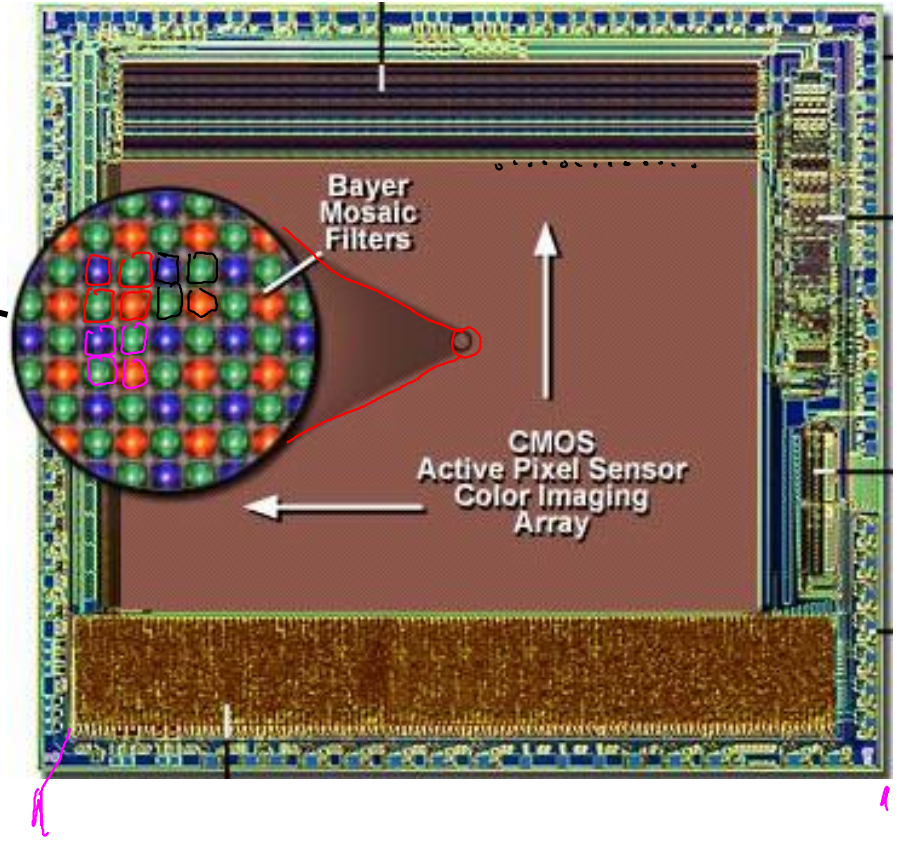
18.ago.2026

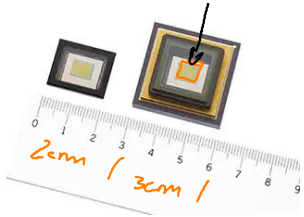
Processamento Digital de Imagens

- Sensor CMOS
  - Color: Bayer

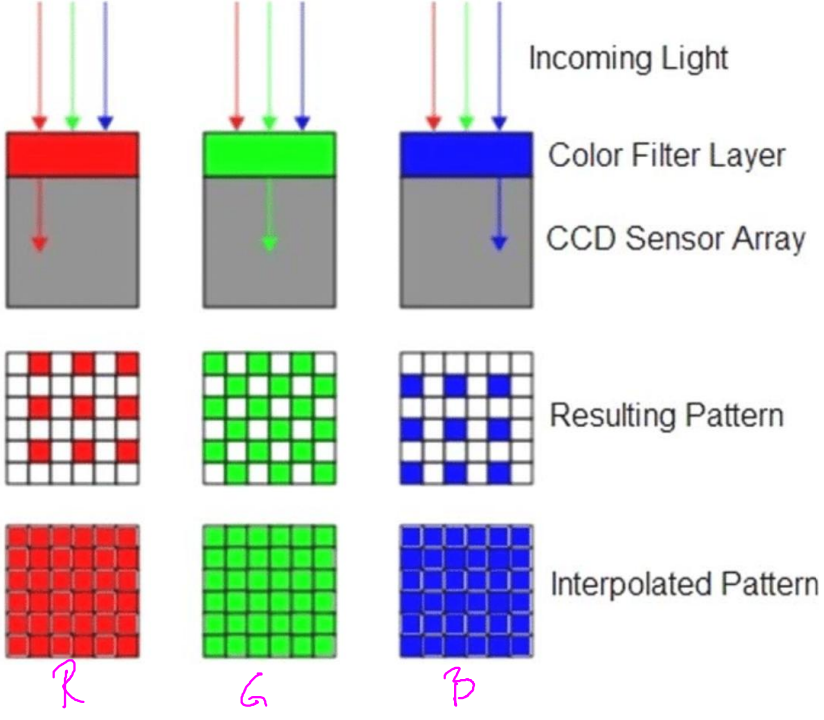


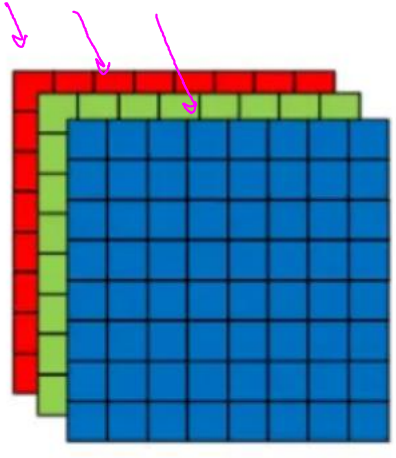
Algoritmo de **demosaicing**, tb chamado de **debayering**





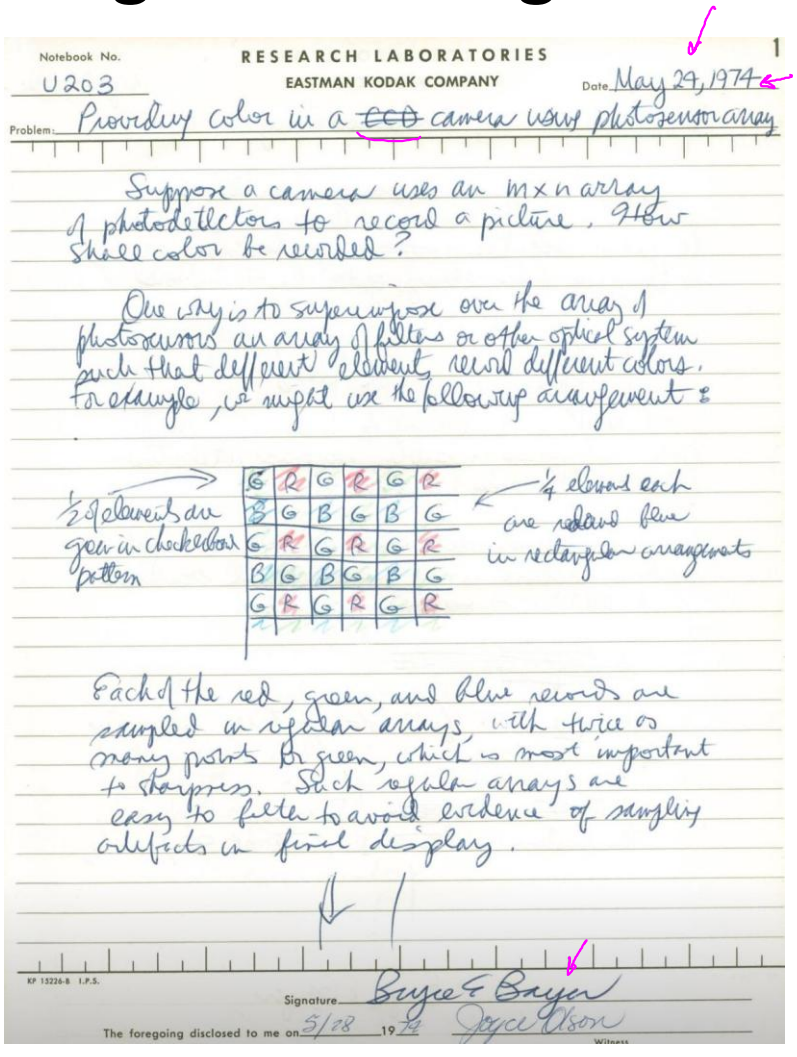
https://www.olympus-lifescience.com/en/microscoperesource/primer/digitalimaging/cmosimagesensors/
https://www.sony.com/en/SonyInfo/News/Press/202005/20-036E/
Outras infos interessantes: https://www.opto-e.com/basics/sensor-characteristics


Incoming Light
Color Filter Layer
CCD Sensor Array
Resulting Pattern
Interpolated Pattern
R
G
B



Processamento Digital de Imagens

- Sensor CMOS
  - Color: Bayer



Notebook No. U203

RESEARCH LABORATORIES
EASTMAN KODAK COMPANY

1

Date May 27, 1974

29/05/1974

Problem: Providing color in a ~~CCD~~ camera using photosensor array

Suppose a camera uses an m x n array of photodetectors to record a picture. How shall color be recorded?

One way is to superimpose over the array of photosensors an array of filters or other optical system such that different elements record different colors. For example, we might use the following arrangement:

½ of elements are green in checkerboard pattern

G	R	G	R	G	R
B	G	B	G	B	G
G	R	G	R	G	R
B	G	B	G	B	G
G	R	G	R	G	R

¼ elements each are red and blue in rectangular arrangements

Each of the red, green, and blue records are sampled in regular arrays, with twice as many points in green, which is most important to sharpness. Such regular arrays are easy to filter to avoid evidence of sampling artifacts in final display.

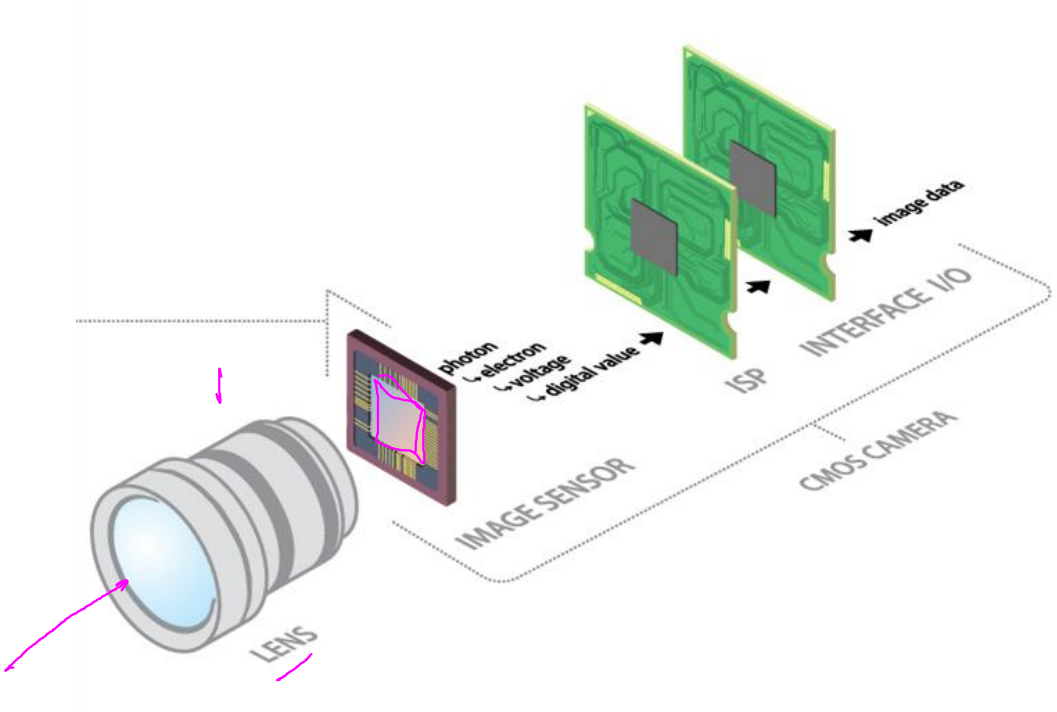
KP 13226-B I.P.S.

Signature Bryce E. Bayer

The foregoing disclosed to me on 5/28 19 74 Witness Joyce Olson

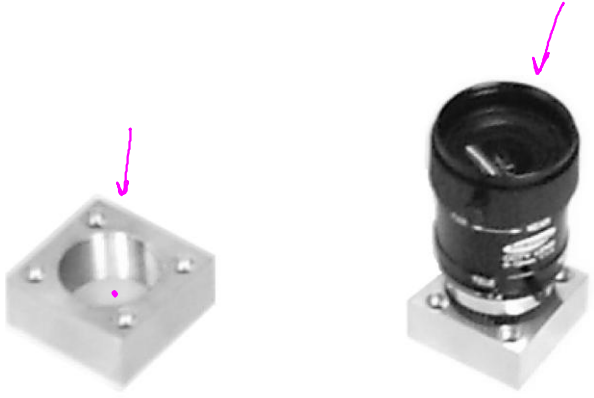
Processamento Digital de Imagens

- Etapas
  - IMPORTANTE: Lentes!



https://thinklucid.com/tech-briefs/understanding-digital-image-sensors/

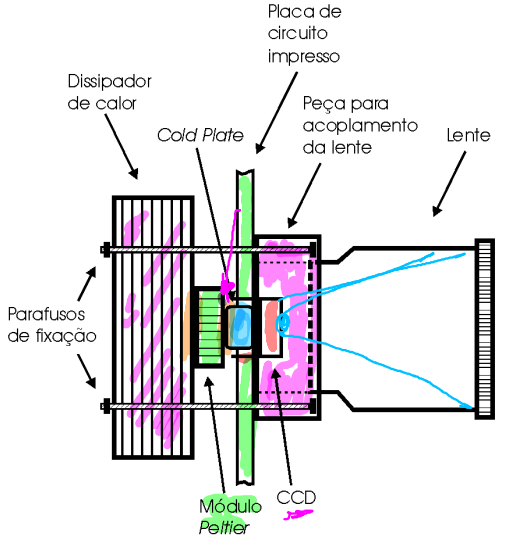
Processamento Digital de Imagens

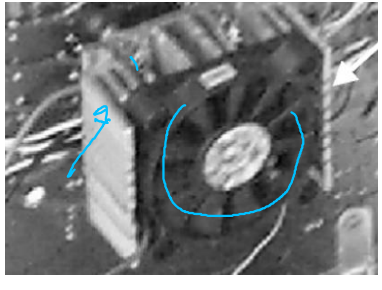


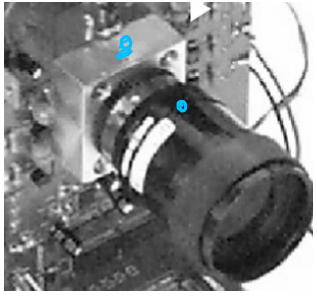
(a) (b)

(a) Foto da peça para acoplamento da lente.
(b) Peça com lente C- *Mount* acoplada.

Montagem do sensor de imagem, lente e módulo de resfriamento *Peltier*.







Processamento Digital de Imagens

- Cálculo da distância focal de uma lente

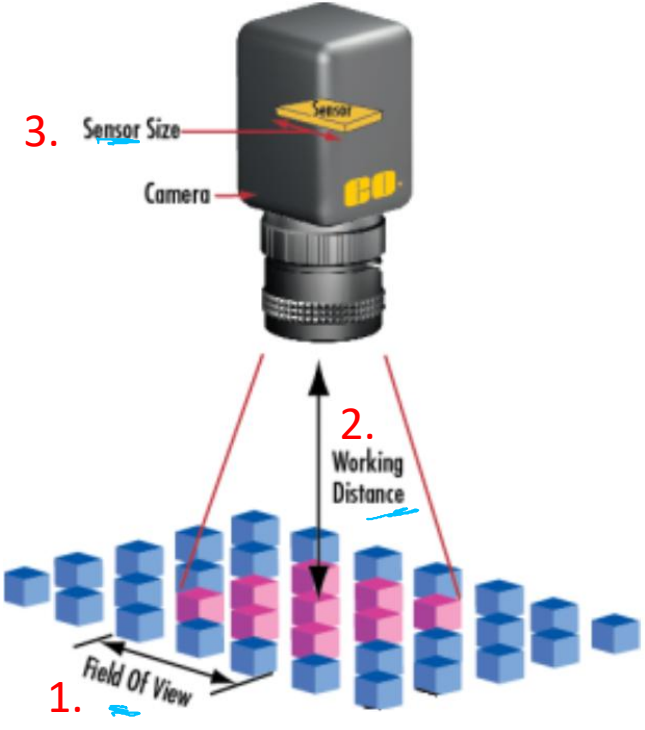
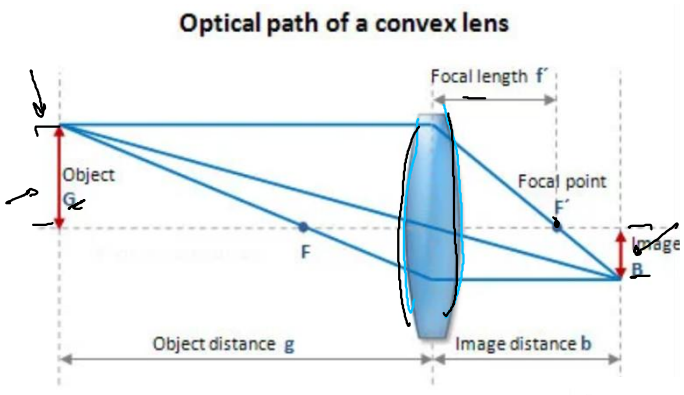


Figura adaptada de https://www.edmundoptics.com/knowledge-center/application-notes/imaging/6-fundamental-parameters-of-an-imaging-system/

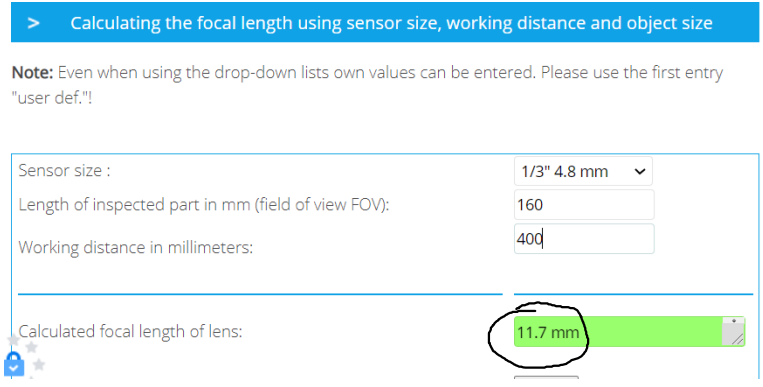
Processamento Digital de Imagens

- Cálculo da distância focal de uma lente
  - **Exemplo**: Uma aplicação requisita um campo de visão de 16 cm, e a distância de trabalho é de 40 cm. O sensor da câmera é de 4,8 mm (1/3”). Que lente preciso usar?



$$f = \frac{g}{\frac{G}{B} + 1}$$

https://www.vision-doctor.com/en/optical-calculations.html





Arducam CS-Mount Lens Kit for Raspberry Pi HQ Camera (Type 1/2.3), 6mm to 25mm Focal Lengths, 65 to 14 Degrees, Telephoto, Wide Angle, Pack of 5

by Arducam

2 ratings

Price: $109.99 + $133.85 Shipping & Import Fees Deposit to Brazil Details

*: CS-Mount Lens Kit

$109.99 | $99.99

Brand	Arducam
Max Focal Length	25 Millimeters
Min Focal Length	6 Millimeters

https://www.amazon.com/Arducam-CS-Mount-Raspberry-Lengths-Telephoto/dp/B088GP9K4T/ref=pb_allspark_session_sims_desktop_421_4/130-9945989-8896219?_encoding=UTF8&pd_rd_i=B088GP9K4T&pd_rd_r=d5f4bbb8-094b-46a4-bbc8-b4e46a2a1f2e&pd_rd_w=qm2M1&pd_rd_wg=kNtTo&pf_rd_p=e500004d-dce5-4973-9afd-bba519c83f08&pf_rd_r=VSGCW7Z6RP0A87SW84WE&psc=1&refRID=VSGCW7Z6RP0A87SW84WE

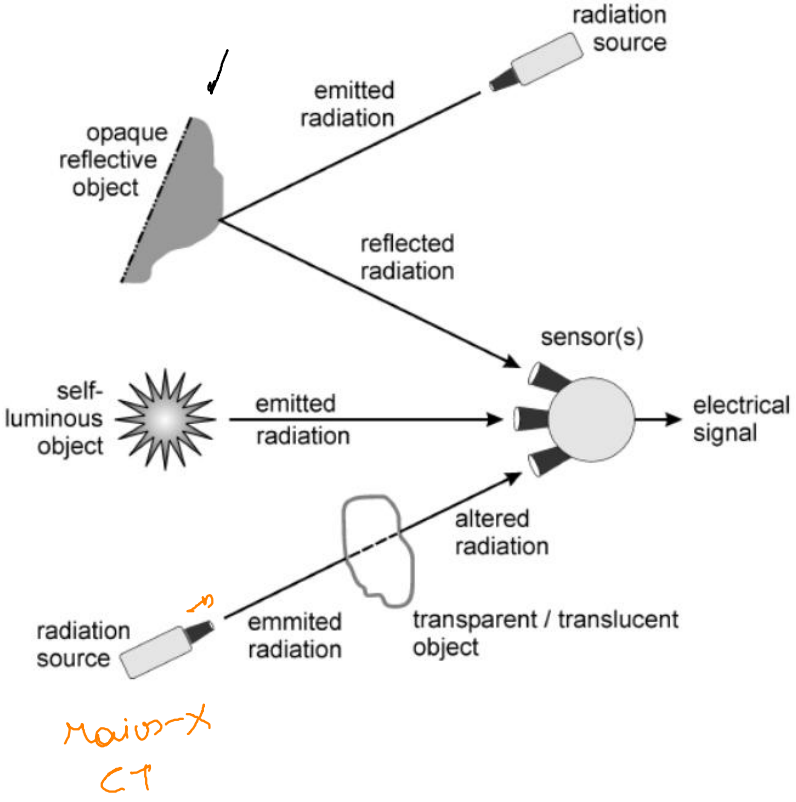
Processamento Digital de Imagens

- Tipos de imagem

Imagem de reflexão

Imagem de emissão

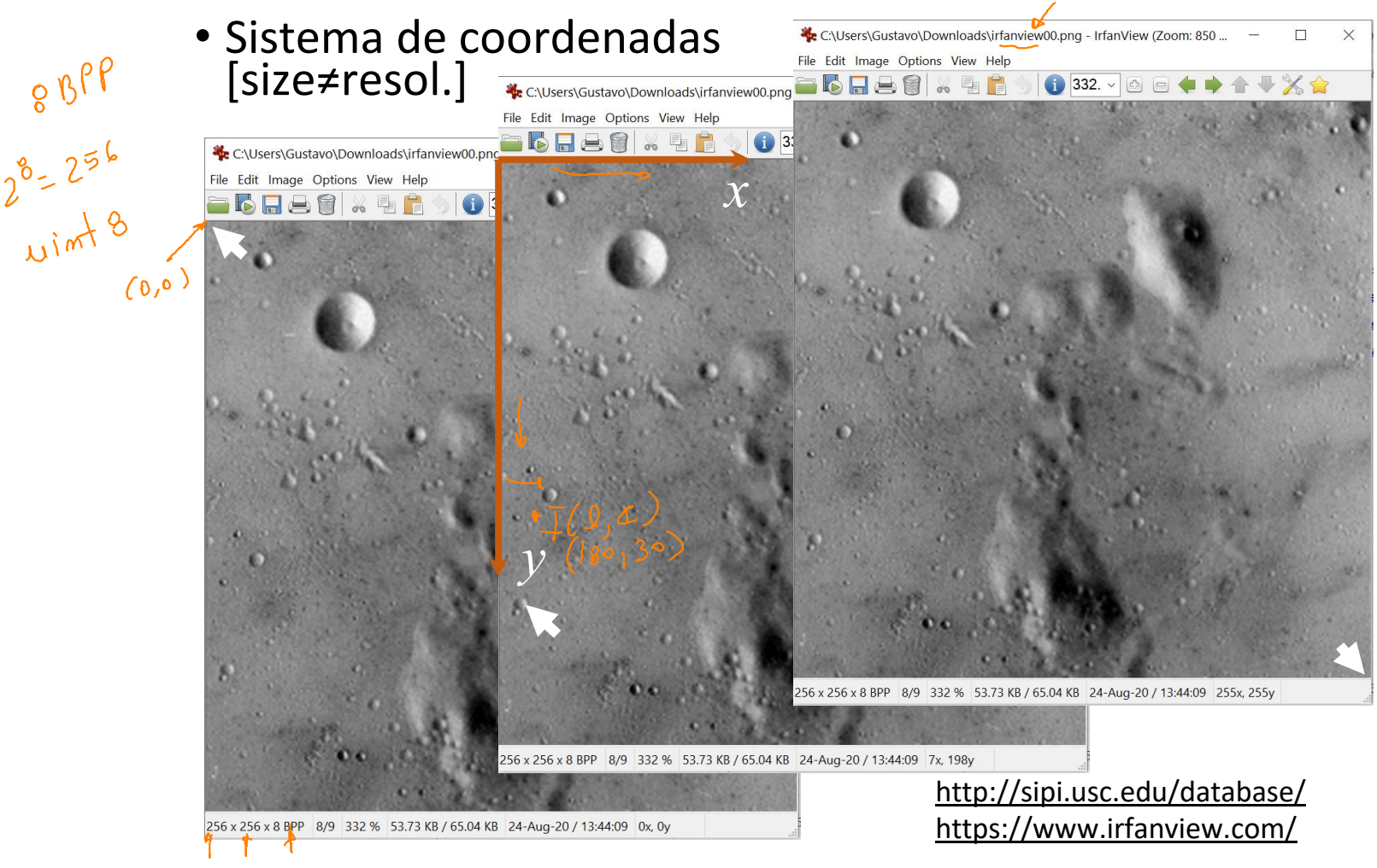
Imagem de absorção



Oge Marques, Practical Image and Video Processing Using MATLAB, John Wiley and Sons, 2011. Figura 5.3

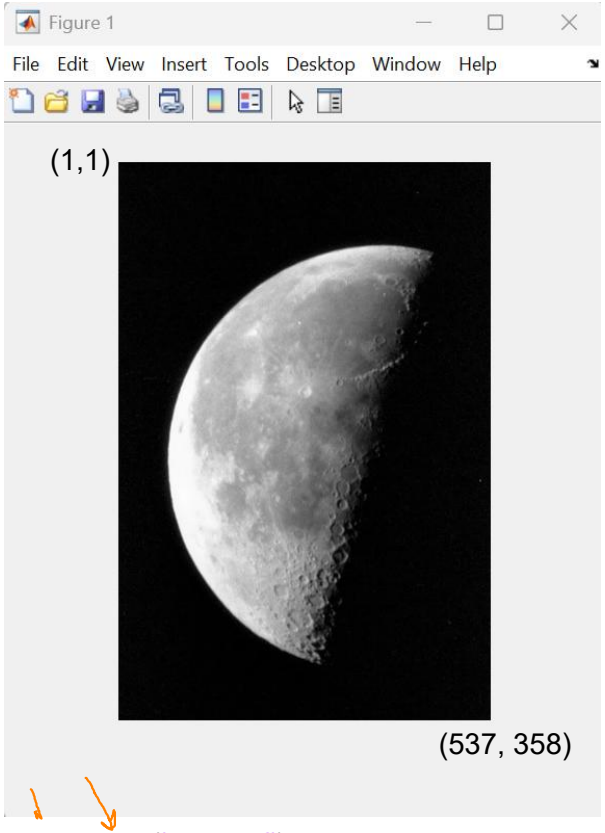
Processamento Digital de Imagens

- Sistema de coordenadas [size≠resol.]

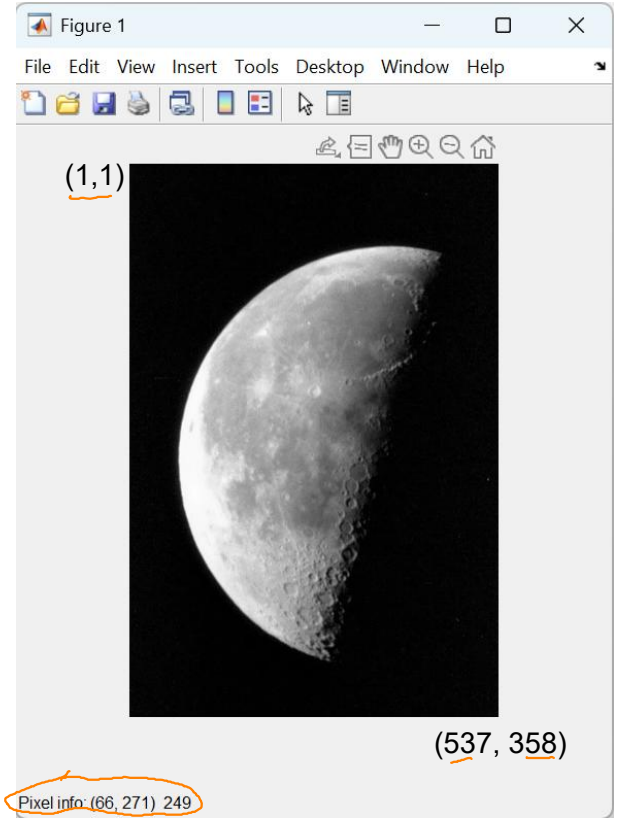


http://sipi.usc.edu/database/
https://www.irfanview.com/

Processamento Digital de Imagens



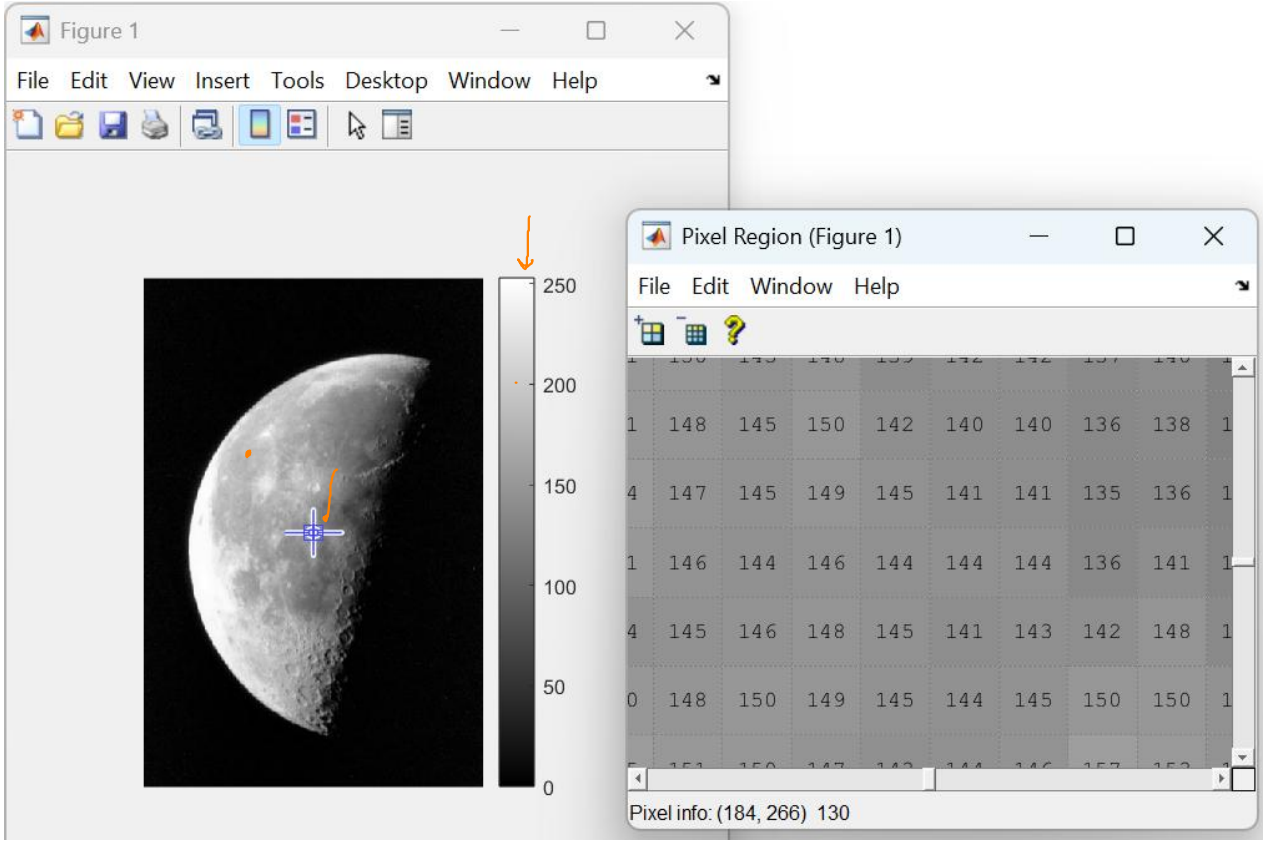

```
I = imread('moon.tif');
imshow(I);
```




```
I = imread('moon.tif');
imshow(I);
impixelinfo;
```

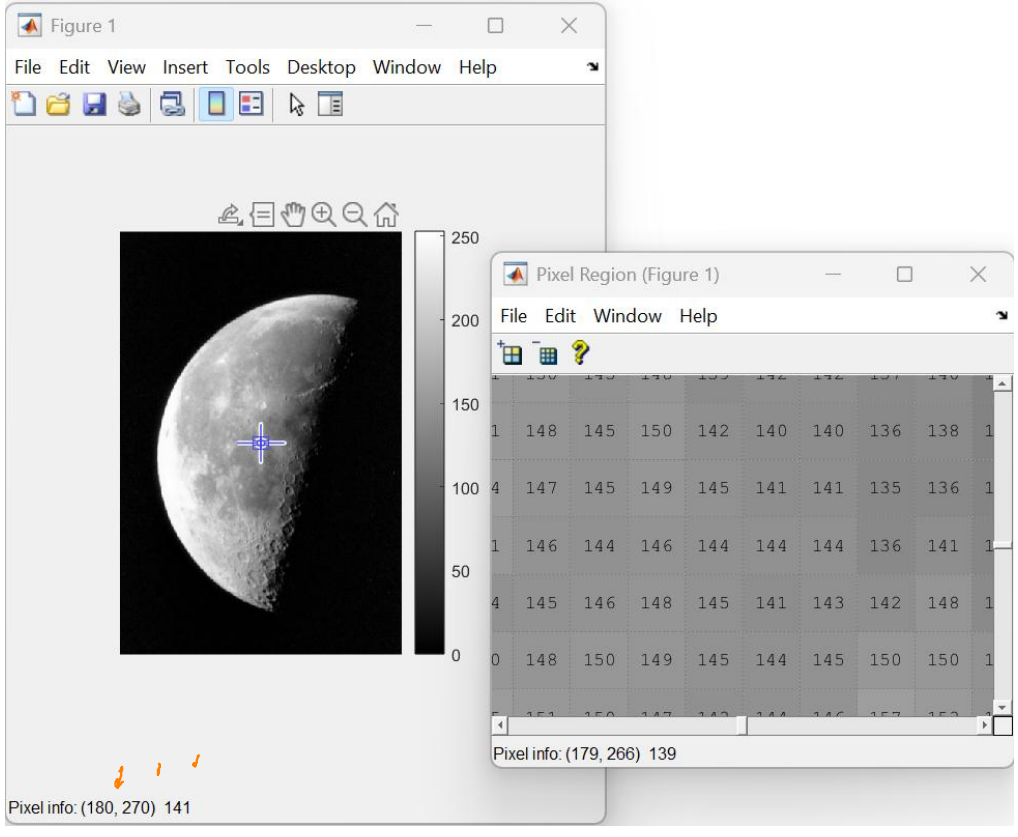
Processamento Digital de Imagens

```
clc, clear, close all
I = imread('moon.tif');
figure
imshow(I,[])
colormap(gray)
colorbar
impixelregion
```



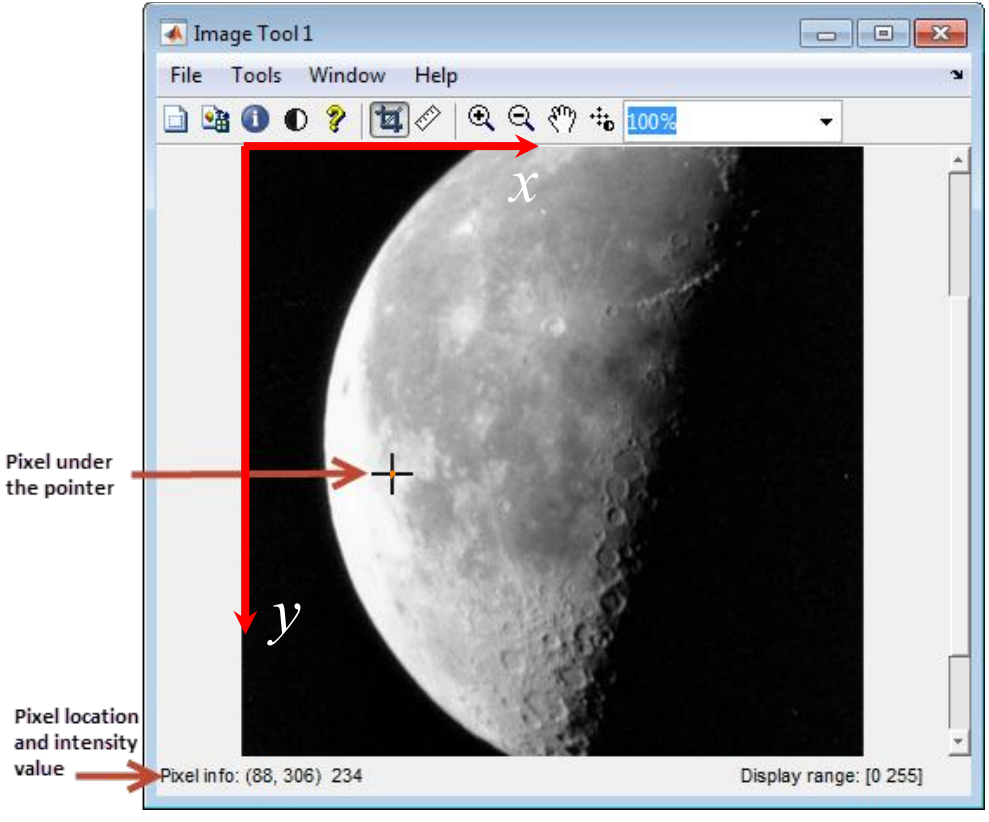
Processamento Digital de Imagens

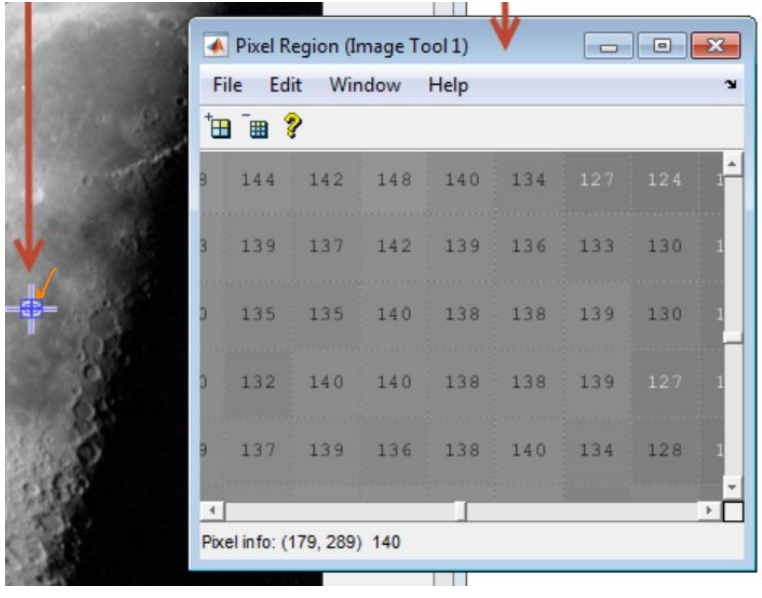
```
clc, clear, close all
I = imread('moon.tif');
figure
imshow(I,[])
colormap(gray)
colorbar
impixelinfo
impixelregion
```



Processamento Digital de Imagens

- Matriz imagem e sistema de coordenadas (grayscale)





https://www.mathworks.com/help/images/get-pixel-information-in-image-viewer-app.html